

## Subject: Biology and Environmental Science (CY202)

### Question Bank

#### Unit 5: Water Pollution Abatement and Solid Waste Management

##### Short questions:

1. What are the three types of water quality parameters. Mention with examples.
2. What do you mean by Total dissolved solids (TDS)? How can TDS be used to measure ionic conductance of water?
3. What is the saturation concentration of Dissolved oxygen (DO) at 20°C and 0°C?
4. Mention any two methods of determination of Dissolved oxygen (DO) in water.
5. The results from a BOD test, diluted by 100, are given in columns 1 and 2 of the Table-1. Calculate BOD<sub>5</sub>.

| Time (days) | Diluted sample DO (mg/L) | BOD (calculated) (mg/L) |
|-------------|--------------------------|-------------------------|
| 0           | 7.95                     | -----                   |
| 1           | 3.75                     | 420                     |
| 3           | 2.75                     | 520                     |
| 5           | 1.80                     | 615                     |

6. What is eutrophication?
7. What is activated sludge?
8. Discuss in brief, the working principle of rotating biological contactor.
9. Mention the goals of circular economy.
10. What are the principles of 3R.
11. Mention four benefits of e-waste recycling.
12. Write any four effects of e-waste.

##### Broad questions:

1. What are the three phases of waste water treatment? Discuss with diagrams, the working principles of trickling filter.
2. What are biological oxidation ponds? Discuss their working principle with proper diagram.
3. What is integrated solid waste management (ISWM)? Explain elaborately the four functional elements of ISWM.
4. Describe the major components of a hazardous waste management plan.
5. What is e-wastes? Cite their common sources and composition. Discuss the steps involved in e-waste recycling.
6. Discuss the benefits of e-waste recycling.
7. What is Electromagnetic radiation (EMR). Mention its two types. Write the impacts of EMR exposure on human health. Explain in brief, the mitigation techniques.